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Example

Consider the following equilibrium reactions and relate their equilibrium, constants

i) $N_2 + O_2 \rightleftharpoons 2NO$; K_1

ii) $2NO + O_2 \rightleftharpoons 2NO_2$; K_2

iii) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; K_3

$$K_1 = \frac{[NO]^2}{[N_2][O_2]}$$

$$K_2 = \frac{[NO_2]^2}{[NO]^2[O_2]}$$

$$\text{Now } K_1 \times K_2 = \frac{[NO]^2}{[N_2][O_2]} \times \frac{[NO_2]^2}{[NO]^2[O_2]}$$

$$= \frac{[NO_2]^2}{[N_2][O_2]^2} = K_3$$

$$\therefore K_3 = K_1 \times K_2$$

8.7.2 Predicting the direction of a reaction

From the knowledge of equilibrium constant, it is possible to predict the direction in which the net reaction is taking place for a given concentration or partial pressure of reactants and products.

Consider a general homogeneous reversible reaction,

$$xA + yB \rightleftharpoons lC + mD$$

For the above reaction under non-equilibrium conditions, reaction quotient 'Q' is defined as the ratio of the product of active masses of reaction products raised to the respective stoichiometric coefficients in the balanced chemical equation to that of the reactants.

Under non-equilibrium conditions, the reaction quotient Q can be calculated using the following expression.

$$Q = \frac{[C]^l [D]^m}{[A]^x [B]^y}$$

As the reaction proceeds, there is a continuous change in the concentration of reactants and products and also the Q value until the reaction reaches the equilibrium. At equilibrium Q is equal to K_c at a particular temperature. Once the equilibrium is attained, there is no change in the Q value. By knowing the Q value, we can predict the direction of the reaction by comparing it with Kc.

- If $Q = K_c$, the reaction is in equilibrium state.
- If $Q > K_c$, the reaction will proceed in the reverse direction i.e., formation of reactants.
- If $Q < K_c$, the reaction will proceed in the forward direction i.e., formation of products.

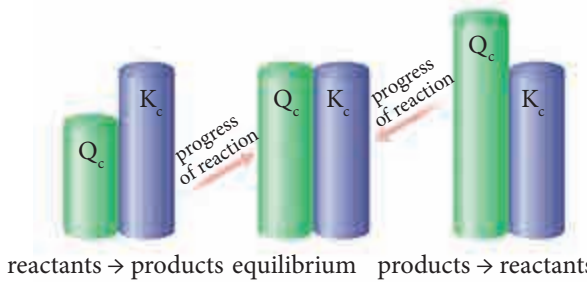


Fig: 8.4 Predicting the direction of a reaction

Example 1

The value of K_c for the following reaction at 717 K is 48.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$